

Frequent Subjective Assessment in Virtual Reality based on Click

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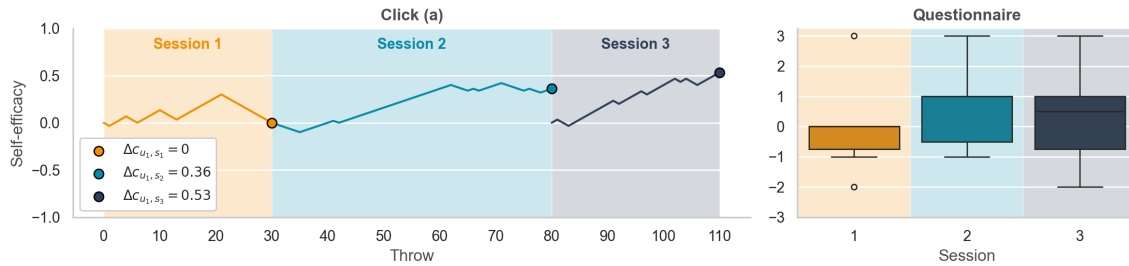


Figure 1: Click-based (a) and questionnaire-based (b) data collected from a single user. Our method captures what happens throughout the experience, rather than relying only on a retrospective median collected at the end of each session.

ABSTRACT

In virtual reality user studies, questionnaires are commonly used to collect subjective data but rely on retrospective reporting, limiting their ability to capture in-experience variations. We propose a method for frequent in-game subjective assessment based on a simple controller button click. As a use case, participants reported their self-efficacy before each shot in a virtual ball-throwing task. Results show a statistically significant positive correlation between the click-based and questionnaire-based measures, indicating the potential of this approach for future virtual reality studies.

Index Terms: Virtual Reality, Assessment, Self-Efficacy, Interaction, User Studies, Human Factors

1 INTRODUCTION AND RELATED WORKS

Many Virtual Reality (VR) studies rely on post-experience questionnaires to gather subjective user data. However, this retrospective approach obscures how participants' perceptions evolve during the experience. Further drawbacks of using questionnaires include memory biases that can reduce response accuracy [4] and, often, the need to remove the VR headset, which can reduce the user's sense of presence (a phenomenon known as Breaks in Presence, or BIP) [7]. Prior works have addressed these issues, embedding surveys directly into the experience itself [6, 4]. Yet, even in-VR questionnaires cannot fully capture quick variations, as they are typically lengthy, and repeating them multiple times during an experiment would be impractical. In the current work, we propose a data-collection method based on a simple, quick action (in this use case, a button click) that enables frequent sampling throughout the experiment and compare it to a post-experience questionnaire to assess whether it can be an effective alternative or complementary tool. Bouchard *et al.* [2] validated a single-item measure for assessing presence; however, while many studies have focused primarily on this construct and BIP, other human factors have received less attention. We therefore focused on self-efficacy, defined as an



Figure 2: Apparatus: HTC Vive Cosmos Elite (HMD), Vive controller (left hand, used for self-efficacy input Yes/No), Vive tracker (right hand, used to throw the ball).

individual's belief in their ability to achieve a certain level of performance [1]. We hypothesize that (H.) frequent collection of subjective data from user input can complement or provide an accurate alternative to post-experience questionnaires.

2 METHODS AND RESULTS

As a use case, we designed a ball-throwing VR scenario [3, 5] where users were asked to throw a virtual ball and hit a target. A total of 30 participants took part in the experiment (23 m, 7 f; age between 20 and 34, $\mu = 25.5$, 17 different nationalities). They completed 3 sessions (s_1, s_2, s_3), each one including a different number of throws ($s_1 : 30; s_2 : 50; s_3 : 30$). Users were divided into three groups, with each group experiencing a different difficulty level in hitting the target in s_2 . Self-efficacy data per user u and per session s were collected through two methods:

1. **Questionnaire:** At the end of each session, users completed a custom questionnaire (Table 1) using a 7-point Likert scale from -3 (not at all) to $+3$ (absolutely). We then computed the median score $q_{u,s}$.
2. **Click:** At the beginning of the experiment, participants were provided with a binary Yes/No question, "Do you think your next shot will be successful?", used to assess their self-efficacy. Periodically (in this use case, before each throw), users answered the question by clicking the buttons assigned

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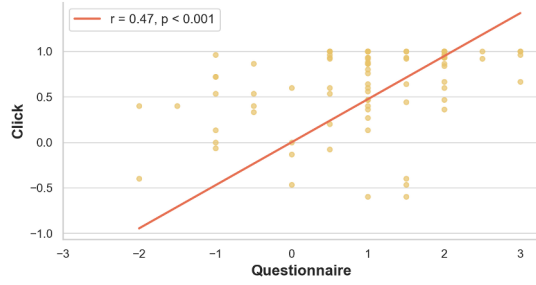


Figure 3: Line plot showing the repeated measures correlation (rmcorr) coefficient (r).

to “Yes” or “No” on the VR controller (Fig. 2). In each session, the click-based self-efficacy score was initialized to 0 and updated by adding +1 (Yes) or -1 (No) after each response. At the end of each session, the final value was normalized by the total number of throws, yielding the change from start to end, denoted as $\Delta c_{u,s}$ (Fig. 1).

We calculated the repeated measures correlation between the click ($\Delta c_{u,s}$) and questionnaire ($q_{u,s}$) scores, which accounts for multiple measurements (one for each session) from each user. Results show a statistically significant, moderate positive correlation ($r = 0.47$, $p < 0.001$, Fig. 3). The 95% confidence interval ($[0.25, 0.65]$) indicates that the true correlation is likely moderate to strong, confirming the reliability of the observed effect. Additionally, the statistical power of **0.98** suggests that the sample size was sufficient to detect this correlation. Figure 4 shows that all participant-specific repeated measures correlation fits have positive slopes, indicating a consistent positive correlation across participants.

Table 1: Self-efficacy questionnaire items.

No.	Question
Q1	If you were asked to increase the success rate in hitting the target, do you think you could achieve it?
Q2	If you were asked to perform the same actions in a real environment instead of VR, do you think you could accomplish the task in the same way?
Q3	Would you be able to handle the task well even if the difficulty level increased (e.g., the target becomes more distant or smaller)?
Q4	Do you feel you could cope well with different types of movements with the same VR interface?
Q5	How difficult did you find this ball throwing task?
Q6	Would you like to try the task again to improve your current success rate?

3 DISCUSSION AND CONCLUSIONS

Questionnaires used in VR experiments, by design, do not capture in-game variations of the perceptions of the users. We therefore proposed a method for frequently collecting subjective data based on a quick action, a button click, enabling capturing changes within the experience (Fig. 1). A repeated measures correlation between this approach and a post-experience questionnaire revealed a statistically significant, moderate-to-strong correlation. This suggests that collecting data via frequent in-game user input could be a promising alternative or complementary method to standard questionnaires, supporting our hypothesis **H**. However, this study has some limitations: i) it focused on a single factor (self-efficacy) and

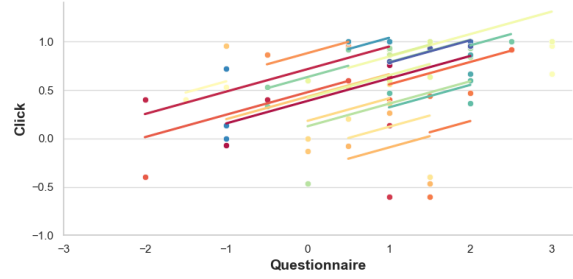


Figure 4: Repeated measures correlation (rmcorr) fits per user; positive slopes indicate positive correlation.

a single task (ball-throwing), while the observed correlation may differ in other cases; ii) the questionnaire lacked prior validation. Further investigation is planned i) to evaluate the generalizability of the method over other human factors (such as sense of embodiment, agency, ownership, affect, enjoyment, fatigue) and ii) to examine whether high-frequency requests for subjective reports influence reflection, attention, engagement, self-efficacy, or performance.

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